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**# PROBLEM 1**

```
X <- c(6,5,8,8,7,6,10,4,9,7)
```

```
Y <- c(8,7,7,10,5,8,10,6,8,6)
```

```
mean_X <- mean(X)
```

```
mean_Y <- mean(Y)
```

```
sx <- sd(X)
```

```
sy <- sd(Y)
```

```
r <- cor(X, Y)
```

```
sy_x <- sy * sqrt(1 - r^2)
```

```
sx_y <- sx * sqrt(1 - r^2)
```

```
cat("-----\n")
```

```
cat("PROBLEM 1 RESULTS\n")
```

```
cat("-----\n")
```

```
cat("Mean of X =", mean_X, "\n")
```

```
cat("Mean of Y =", mean_Y, "\n")
```

```
cat("Standard deviation of X =", sx, "\n")
```

```
cat("Standard deviation of Y =", sy, "\n")
```

```
cat("Correlation r =", r, "\n")
```

```
cat("s_y.x =", sy_x, "\n")
```

```
cat("s_x.y =", sx_y, "\n\n")
```

**# PROBLEM 2**

```
data(pressure)
```

```
X <- pressure$temperature
```

```
Y <- pressure$pressure
```

```
r_original <- cor(X, Y)
```

```
X_new <- 2*X + 6
```

```
Y_new <- 3*Y - 15
```

```
r_new <- cor(X_new, Y_new)
```

```
cat("-----\n")
```

```
cat("PROBLEM 2 RESULTS\n")
cat("-----\n")
cat("Correlation (original) =", r_original, "\n")
cat("Correlation (transformed) =", r_new, "\n\n")
```

```
-----
PROBLEM 1 RESULTS
-----
Mean of X = 7
Mean of Y = 7.5
Standard deviation of X = 1.825742
Standard deviation of Y = 1.649916
Correlation r = 0.5532833
s_y.x = 1.374369
s_x.y = 1.52083

-----
PROBLEM 2 RESULTS
-----
Correlation (original) = 0.7577923
Correlation (transformed) = 0.7577923
```

The correlation is remains the same after transformation

**Note:**

***For Problems 3 and 4, I used a subset of the dataset instead of the full package-based dataset due to limitations in installing the necessary packages in my laptop (chromebook). This alternative approach was used to answer and complete the analysis.***

**# PROBLEM 3**

```
my_data <- data.frame(
  arr_delay = c(28, 48, 19, 10, 28, 0, 14, 37, 60, 15, 18, -1, -2, 17, 44, 32, 38, 19, -11, 20),
  dep_time = c(2333, 32, 1147, 754, 2322, 1023, 1836, 1634, 743, 239, 221, 26, 434, 900, 634,
  836, 646, 813, 702, 1100),
  dep_delay = c(28, 41, 18, 4, 32, -3, 10, 28, 47, 8, 12, 0, 0, 13, 42, 25, 29, 13, -8, 11),
```

```

distance = c(969, 1715, 1415, 1155, 1384, 659, 221, 947, 674, 1282, 2247, 1175, 2006, 389,
762, 2099, 1467, 1728, 846, 2268),
air_time = c(113, 205, 171, 145, 165, 79, 19, 106, 90, 170, 266, 145, 229, 43, 87, 267, 179,
193, 80, 262),
month     = c(4, 8, 7, 12, 9, 8, 5, 2, 5, 8, 10, 9, 12, 12, 12, 9, 1, 9, 7, 9),
hour      = c(23, 0, 11, 7, 23, 10, 18, 16, 7, 2, 2, 0, 4, 9, 6, 8, 6, 8, 7, 11)
)

```

```

model <- lm(arr_delay ~ dep_time + dep_delay + distance + air_time + month + hour, data =
my_data)

```

```

summary(model)

```

```

Call:
lm(formula = arr_delay ~ dep_time + dep_delay + distance + air_time +
    month + hour, data = my_data)

Residuals:
    Min       1Q   Median       3Q      Max
-6.1526 -2.1038 -0.1037  3.1032  5.6912

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) 10.80756    5.47536   1.974   0.070 .
dep_time     -0.05808    0.08328  -0.697   0.498
dep_delay     1.09009    0.06839  15.940 6.51e-10 ***
distance     -0.01798    0.01636  -1.099   0.292
air_time      0.15462    0.13570   1.139   0.275
month        -0.60387    0.34983  -1.726   0.108
hour          5.55565    8.31941   0.668   0.516
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.185 on 13 degrees of freedom
Multiple R-squared:  0.9636,    Adjusted R-squared:  0.9468
F-statistic: 57.34 on 6 and 13 DF,  p-value: 1.33e-08

```

## Problem 4

### # PROBLEM 4

*# Arrival Delay*

```
Y <- c(15, -8, 55, -2, 12, -10, 40, 105, 5, 22)
```

```
X1 <- c(1, 0, 1, 0, 1, 0, 1, 1, 1, 1)
```

*# Departure Delay in minutes*

```
X2 <- c(20, -5, 60, 0, 15, -3, 45, 110, 8, 25)
```

*# Air Time in minutes*

```
X3 <- c(120, 85, 250, 45, 180, 60, 210, 320, 115, 140)
```

*# Distance in miles*

```
X4 <- c(800, 500, 1600, 200, 1100, 350, 1450, 2200, 750, 950)
```

*# Hour of the day*

```
X5 <- c(8, 11, 15, 7, 18, 6, 20, 22, 10, 13)
```

```
X6 <- c(1, 0, 1, 0, 1, 0, 1, 1, 0, 1)
```

```
flight_data <- data.frame(Y, X1, X2, X3, X4, X5, X6)
```

```
model <- lm(Y ~ X1 + X2 + X3 + X4 + X5 + X6, data = flight_data)
```

```
summary(model)
```

Call:

```
lm(formula = Y ~ X1 + X2 + X3 + X4 + X5 + X6, data = flight_data)
```

Residuals:

|  | 1          | 2         | 3         | 4          | 5          | 6          | 7          |
|--|------------|-----------|-----------|------------|------------|------------|------------|
|  | 1.081e-01  | 1.568e+00 | 3.210e-01 | -5.038e-02 | -6.392e-01 | -1.518e+00 | -6.625e-01 |
|  | 8          | 9         | 10        |            |            |            |            |
|  | -6.451e-03 | 2.671e-16 | 8.791e-01 |            |            |            |            |

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t ) |     |
|-------------|----------|------------|---------|----------|-----|
| (Intercept) | -4.88046 | 1.72503    | -2.829  | 0.066233 | .   |
| X1          | 4.42575  | 2.38685    | 1.854   | 0.160749 |     |
| X2          | 1.11919  | 0.07699    | 14.536  | 0.000706 | *** |
| X3          | 0.10385  | 0.07850    | 1.323   | 0.277662 |     |
| X4          | -0.02790 | 0.01501    | -1.858  | 0.160123 |     |
| X5          | 0.54815  | 0.26404    | 2.076   | 0.129493 |     |
| X6          | -1.56646 | 1.80213    | -0.869  | 0.448679 |     |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.472 on 3 degrees of freedom

Multiple R-squared: 0.9994, Adjusted R-squared: 0.9983

F-statistic: 862.5 on 6 and 3 DF, p-value: 6.097e-05